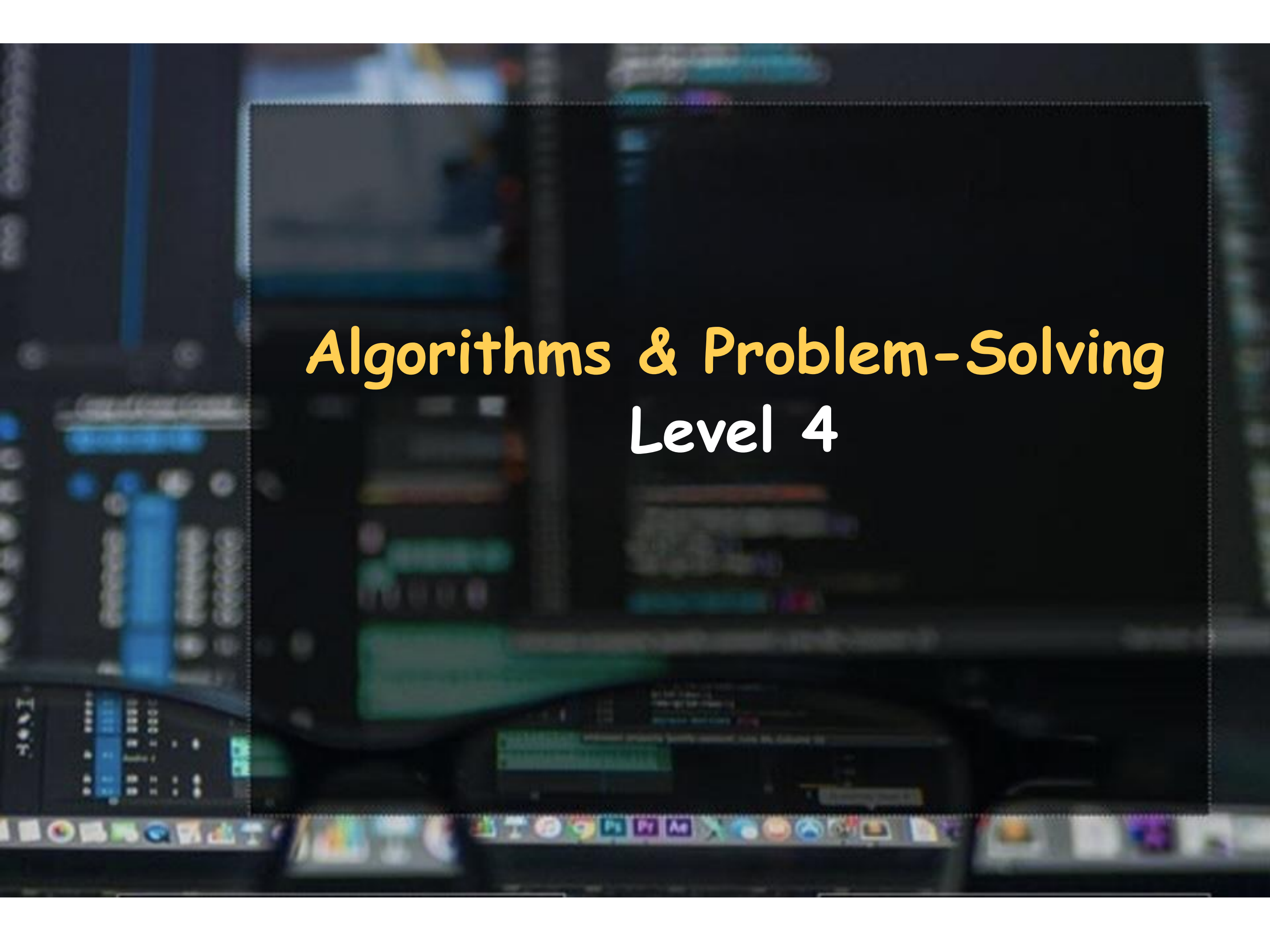
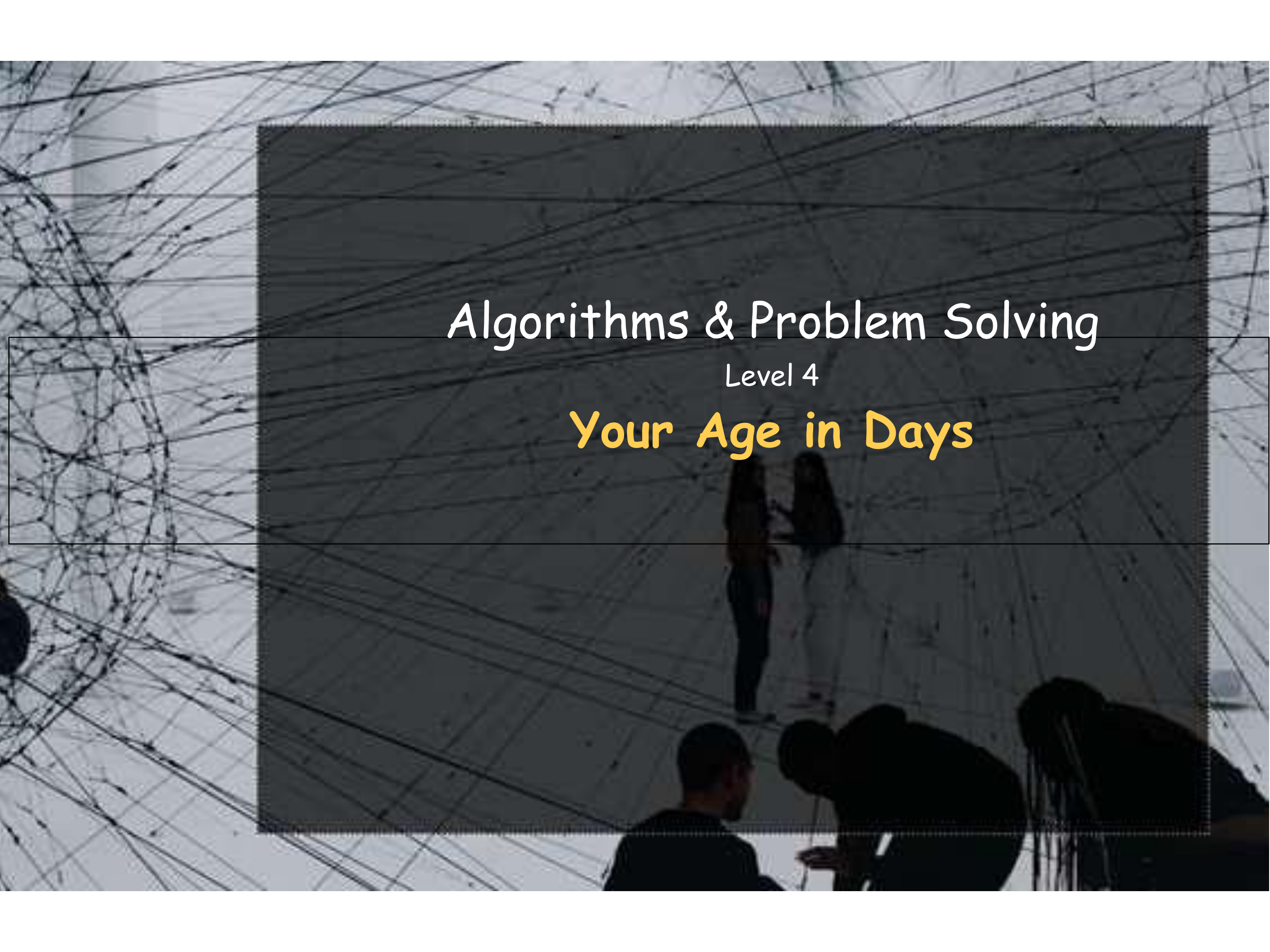




Algorithms & Problem-Solving Level 4

The background of the slide is a complex network diagram with numerous nodes and edges. A large, dark, semi-transparent rectangle is centered on the slide, serving as a backdrop for the text. The text is white and yellow, providing high contrast against the dark background.

Algorithms & Problem Solving

Level 4

Your Age in Days



Definition

- The goal is a program that calculates the user's age in days based on their birth date and the current date.

Procedures

- 1 Defining the `sDate` structure:
 - It contains three values: `Year`, `Month`, and `Day` to represent the date.
- 2 Reading the date from the user:
 - Three functions to read the day, month, and year (`ReadDay`, `ReadMonth`, and `ReadYear`).
 - These are then used inside `ReadFullDate` to generate a full date.
- 3 Checking for leap years:
 - Using the `IsLeapYear(year)` function, which returns `true` if the year is a leap year.
- 4 Determining the number of days in a month:
 - Using the `NumberOfDaysInAMonth(month, year)` function, which takes February into account (28 or 29 days).
- 5 Checking the last day or last month:
 - `IsLastDayInMonth(date)` and `IsLastMonthInYear(month)` help with correct transitions between months and years.
- 6 Increasing a date by one day:
 - The `IncreaseDateByOneDay(date)` function handles different cases (end of month or end of year).
- 7 Comparing two dates:
 - `IsDate1BeforeDate2(date1, date2)` to find out which date is earlier.

- 8 Calculating the difference between two dates:
 - `GetDifferenceInDays(date1, date2, includeEndDay)` calculates the number of days between two dates. Birthdate and Today's Date
- 9 Get the Current System Date:
 - Use `GetSystemDate()` to fetch the date from the system clock
- 10 Main Program:
 - Reads the birthdate
 - Fetches the current date
 - Displays the result in days



Example:

Please Enter Your Date of Birth:

Please enter a Day? 20

Please enter a Month? 5

Please enter a Year? 1981

Your Age is : 16233 Day(s).

Summary

- ✓ The program calculates the daily difference between two dates.
- ✓ It uses the concept of daily addition and leap dates to ensure high accuracy.
- ✓ Useful for time-related applications such as age calculations or calculating work periods.

Thank you 😊